DESIGN AND DEVELOPMENT OF STEERING SYSTEM FOR FSAE CAR

Abstract - This paper is focused on the design, optimization, and analysis of steering components and the "Design of steering geometry for formula student cars. The main objective of the research is to ensure more rational and effective steering input or response between the driver and the wheels, reducing the amount of work the driver must do and enhancing their engagement with the wheels. The focus of the project is to consider the Ackerman set-up, steering effort, steering arm length, rack motion, turning radius, steering ratio, slip angle, castor, toe angles, king-pin angle, and camber angle in order to obtain that sensitivity. Rack and Pinion serve as the link in this instance between the driver and wheels.
Key Words: Steering, Ackerman, Rack and Pinion, Optimization.

1. INTRODUCTION

The **steering system** is a critical control component of a vehicle, and its geometry plays a vital role in handling and maneuverability. Common geometries include **Ackermann**, **anti-Ackermann**, and **parallel steering**, each with specific advantages depending on vehicle type and operating conditions. For effective turning without tire scrub, all tires must rotate about a common center. Among these, **Ackermann geometry** is preferred in Formula cars for **low-speed cornering** due to its high maneuverability and minimal scrub effect, where the **inner wheel turns more sharply than the outer**. Most modern vehicles use a **rack and pinion** system to convert steering wheel rotation into linear motion. In FSAE cars, the steering system is designed as per rulebook constraints, often using a **trapezoidal arrangement** for compactness. **Tie rod optimization** is essential to minimize wheel slip angles, enhancing cornering performance. The goal is to achieve **100% Ackermann geometry**, reducing the **turning radius** and improving the car's dynamic ability through careful design and simulation.

2. LITERATURE REVIEW

In the design of our FSAE vehicle, **Ackermann steering geometry** was selected due to its superior **low-speed cornering**, reduced **tire scrub**, and improved **maneuverability**. It allows all wheels to rotate about a common center, minimizing slip and enhancing control during tight turns—ideal for events like autocross. For the development process, **SolidWorks** was used for precise 3D modeling and packaging of components, ANSYS for structural analysis and stress validation, and **Lotus Shark** for simulating dynamic behavior and optimizing steering response. This combination ensures that the steering system is not only well-packaged and lightweight, but also structurally reliable and performance-oriented.

3. COMPONENTS IN STEERING SYSTEM

3.1. Rack and pinion:

- To transform rotation into linear motion, rack and pinion gears are used.
- The gear is called the pinion, and the flat, toothed component is called the rack.
- There are two gears in a rack and pinion gear set.
- The rack may be straight or flat, and the pinion gear is a typical round gear.
- The rack's teeth and the pinion gear's teeth fit together perfectly.

3.2. Universal joint:

- Universal joint, commonly referred to as a U-joint, is a kind of mechanical connection that enables two shafts to be coupled and transmit torque while still being able to freely rotate and travel in separate directions.
- When the relative orientation of the two shafts may alter, this is helpful.

3.3. Shaft:

- Shafts are typically circular in cross-section.

- A shaft is a rotating machine part that transfers energy from one part to another or from a machine that produces energy to a machine that consumes energy.

3.4. Quick release:

- The rapid demounting tool has a straightforward design and transfers torque through involute splines.

3.5. Tie rod:

- When the steering wheel is turned, the front wheel pivots on the steering knuckles, which are connected to the steering rack at either end by tie rods.
- The tie rod includes two threaded portions that can be adjusted in length to align the front wheels.

3.6. Steering arm:

- The steering arm is used to connect the tie rod with the upright of the wheel.

4. PROBLEMS FACED

Length of the tie rod plays a critical role in the steering system, improper tie rod length causes toe-in and toe-out of the wheels that causes wear of tires, bump steer of the vehicle, vibration on the steering wheel.

5. SOLUTION CONCEPT

Based on the assumed values for the basic dimension of the wheel track and wheelbase of the vehicle, a geometry is drawn on the solid works software and the Ackerman angle is found. Then the model is simulated on the Lotus Shark software to find the exact tie rod length.

6. METHODOLOGY

i. Problem selection.
ii. Solution concept.
iii. Design objectives.
iv. Simulation and Validation.
v. Material selection.
vi. Analysis. vii. Manufacturing and implementation.

7. DESIGN CALCULATION

★ **Iteration 1:**

Let's assume some Parameters:

Wheelbase	1550 mm
Front Track width	1200 mm
Rear Track width	1150 mm
Turning radius	1978.87 mm
Inner wheel turning angle	39.19 degree
Outer wheel turning angle	26.45 degree

- **Ackermann percentage calculation:**

Ackermann angle = $tan^{-1}\left(\frac{Wheelbase}{Turning\ Radius - \frac{Track\ width}{2}}\right)$

$$\theta_{ack} = tan^{-1}\left(\frac{1550}{1978.87 - \frac{1200}{2}}\right)$$

$$= 48.51 \text{ degree}$$

%Ackermann = $\frac{Ackermann\ angle}{Inner\ wheel\ angle} \times 100$

$$= \frac{48.51\ degree}{39.19\ degree} \times 100$$

$$= 123.7\ \%$$

Interpretation:
The calculated Ackermann percentage exceeds 100%, indicating an over-Ackermann steering geometry. This suggests that the inner wheel is turning more than ideally required, which may lead to a tendency toward oversteer during cornering.

★ **Iteration 2:**
In Iteration 1, the system exhibited an over-Ackermann condition, resulting in a tendency toward oversteering. To improve steering response and reduce excessive inner wheel angle during cornering, we now move to **Iteration 2**. In this step, our objective is to reduce the Ackermann percentage to bring it closer to an optimal value, ensuring better

handling characteristics and improved tire performance.

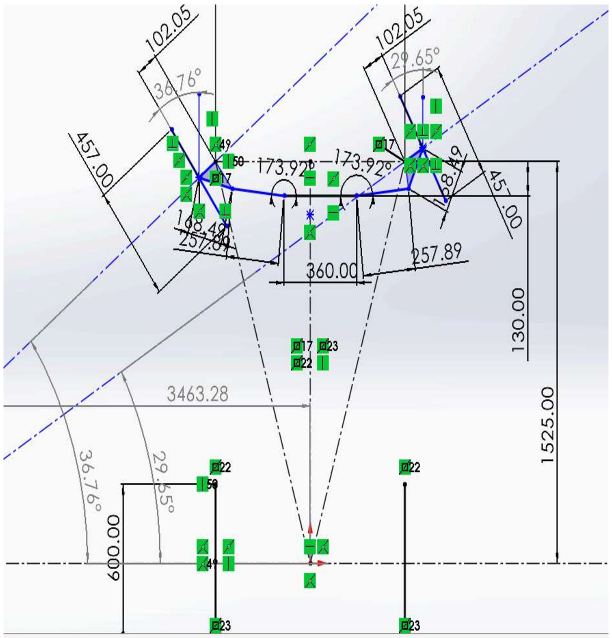


Figure 7.1(Steering geometry 2D sketch)

- **Selected parameters:**

Wheelbase	1525mm
Front Track width	1142mm
Rear Track width	1140mm
Turning radius	3464 mm
Inner wheel turning angle	36.76 degree
Outer wheel turning angle	29.65 degree

- **Ackermann percentage calculation:**

Ackermann angle = $tan^{-1}\left(\frac{Wheelbase}{Turning\ Radius-\frac{Track\ width}{2}}\right)$

$$\theta_{ack} = tan^{-1}\left(\frac{1525}{3464-\frac{1142}{2}}\right)$$

$$= tan^{-1}\left(\frac{1525}{2893}\right)$$

$$\boxed{\theta_{ack} = 27.79\ degree}$$

%Ackermann = $\frac{Ackermann\ angle}{Inner\ wheel\ angle} \times 100$

$$= \frac{27.79}{36.76} \times 100$$

$$\boxed{\%Ackermann = 75.5\%}$$

- **Steering Ratio:**

The ratio of steering wheel turn angle with respect to the maximum turning angle of the front wheel.

S.R. = $\frac{118.4\ degree}{36.76\ degree}$

$$\boxed{S.R. = 3.2}$$

- **Rack Travel:**

The amount of rack travel with respect to the steering ratio needs to be fixed.

The steering wheel radius is 120 mm.

For one complete rotation the steering wheel travel
= 2π x r

= 0.754 m

At one complete rotation of the steering wheel the maximum rack travel is reached with respect to the maximum steer angle.

Steering ratio = $\frac{Steering\ wheel\ travel}{Rack\ travel}$

Rack travel = $\frac{0.754}{3.2}$

Rack travel = 236 mm

Required rack travel is 236 mm.

- **Upper Steering Geometry Design:**

➔ Selection of U-Joint Configuration:

1.Introduction: The upper steering system is crucial for translating driver input to the steering gear. This report focuses on the mathematical analysis of single and double universal (U) joint configurations, justifying the selection of the latter for performance applications like Formula SAE.

2. Steering System Configurations & Calculations:

2.1 Single Universal Joint Configuration(Iteration 1):A single U-joint connects the steering column to the pinion. Its primary issue is non-uniform angular velocity transmission.

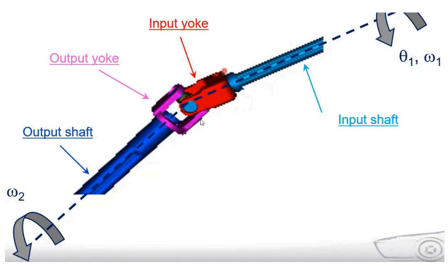


FIg.(Single U-Joint)

- **Operating Angle (α):**
  The angle between the input and output shafts.

- **Input Angular Velocity (ω_1):**
  Constant rotational speed from the steering wheel.

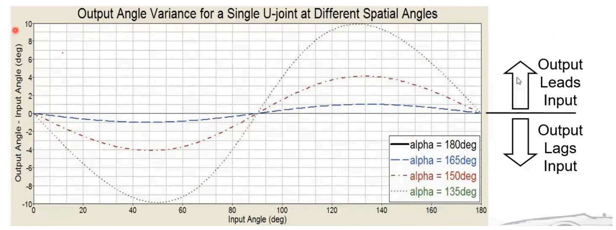
Calculations:

1.Output Angular Velocity (ω_2):

The relationship between input and output angular velocity for a single U-joint is:

$$\omega_2 = \frac{\omega_1 cos(\alpha)}{1 - sin^2(\alpha) cos^2(\theta_1)}$$

Observation: ω_2 is not constant; it varies sinusoidally with θ_1.



2.Output Angular Acceleration (α_2):

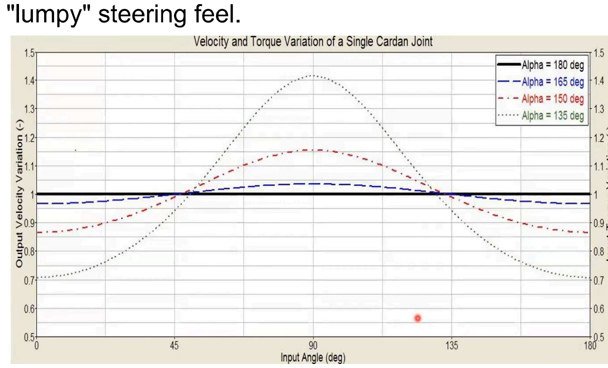
Differentiating ω_2 with respect to time (or θ_1), we find that a non-zero angular acceleration exists even with constant input velocity, leading to dynamic loading and vibration.

3.Torque Ratio:

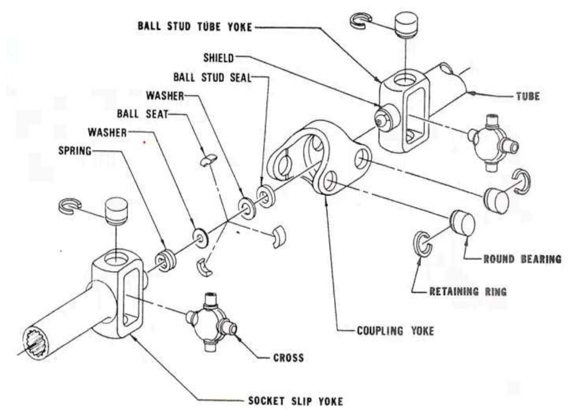
Assuming negligible friction, the torque ratio is the inverse of the velocity ratio:

$$\frac{T_2}{T_1} = \frac{\omega_1}{\omega_2} = \frac{1 - sin^2(\alpha) cos^2(\theta)}{\omega_1 cos(\alpha)}$$

Observation: Input torque required for a constant output torque will fluctuate significantly, resulting in a "lumpy" steering feel.



2.2 Double Universal Joint Configuration (Iteration 2): This configuration uses two U-joints connected by an intermediate shaft. It's designed to achieve uniform angular velocity transmission.



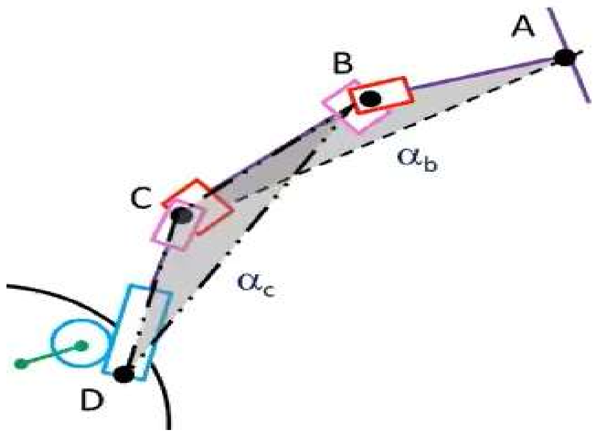
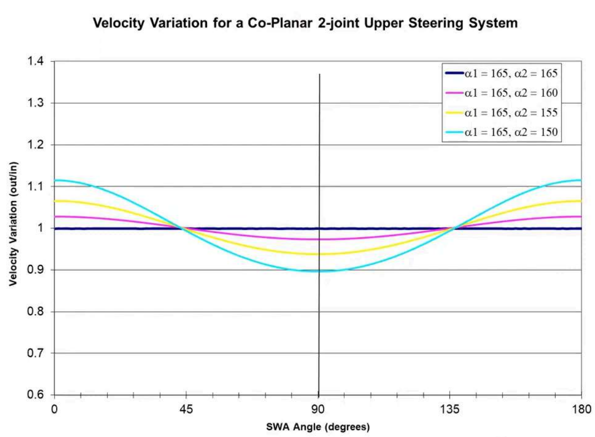


Fig. (Double U-joint)

❖ **Conditions for Uniform Velocity Transmission:**

1.Equal Operating Angles: To achieve zero velocity and Torque variation of the co-planar system the angle of the first U-joint (α_b) must be equal to the angle of the second U-joint (α_c).

$$\alpha_b = \alpha_c$$



2.Phased Orientation: The yokes (forks) on the intermediate shaft must be in phase (parallel to each other). This means if the first joint introduces an acceleration, the second joint, being oriented identically and at the same angle, introduces an equal and opposite deceleration.

Calculations (Under Ideal Conditions):
If the two conditions above are met:

1.Intermediate Shaft Output Velocity ($\omega_{intermediate}$):

$$\omega_{intermediate} = \frac{\omega_1 cos(\alpha_b)}{1 - sin^2(\alpha_b) cos^2(\theta_1)}$$

2.Final Output Velocity(ω_2):

$$\omega_2 = \frac{\omega_{intermediate} cos(\alpha_c)}{1 - sin^2(\alpha_c) cos^2(\phi_{intermediate})}$$

Where $\phi_{intermediate}$ is the input angle to the second joint.
Because the yokes are in phase, $\phi_{intermediate}$ is directly related to θ_1 in a way that cancels the variations.
Therefore, under ideal conditions:

$$\omega_2 = \omega_1$$

3.Torque Ratio:

$$\frac{T_2}{T_1} = \frac{\omega_1}{\omega_2} = 1$$

Observation: This indicates a constant torque ratio, meaning a linear and consistent steering feel.

Justification for Double U-Joint Selection:
The double U-joint is chosen because it inherently supports key steering system requirements:

1.Durability: Enables more accurate sizing calculations for shafts and fasteners by providing uniform motion, leading to less fatigue and failure under "repeated cycling."

2.Stiffness: Allows the system to "maximize the stiffness of components" by minimizing internal compliances due to its uniform velocity transmission, ensuring direct power transfer to the wheels.

3.Smoothness: Achieves uniform motion, ensuring "every newton-meter of steering wheel torque goes towards turning the steering gear pinion," not overcoming internal friction.

4.Comfort: Offers design flexibility to "consider the angle of the steering column off the horizontal" for optimal driver ergonomics and gauge visibility without compromising performance.

❖ **STEERING WHEEL DESIGN AND TORQUE CALCULATIONS:** The steering wheel used in the car is in near oval shape. Thus rotating the steering wheel will be easy to rotate and improves the feel for the driver. The steering wheel was redesigned in order to reduce weight and provide better legroom and visibility. The torque required to rotate the steering wheel by the driver is called Steering Wheel Torque (W_T).

$$W_T = \frac{Kingpin\,Torque(T)}{Steering\ Ratio(S.R)}$$

$$T = W.U.\sqrt{\frac{B^2}{8} + E^2}$$

Where:

W = Load per wheel = 551.8125 N
U = Coefficient of friction = 1.4
E = kingpin offset = 62 mm
B = Width of tire = 8.46 inch

Now,

$$T = 55.8125 \times 1.4.\sqrt{\frac{(8.46)^2}{8} + (62)^2}$$

T = 47.97 mm

Now,

$$W_T = \frac{47.97}{3.2}$$

$$\boxed{W_T \quad = \quad 14.99\ Nm}$$

FORCE applied on steering wheel is = $\frac{W_T}{Radius\ of\ Steering\ wheel}$ = $\frac{14.99\ Nm}{120\ mm}$ = 124.92 N

➔ Steering Effort Estimation and Manual Input Justification:

1.Steering effort is within acceptable limits: Max ~15 Nm torque at wheel is manageable without power assist for most drivers.

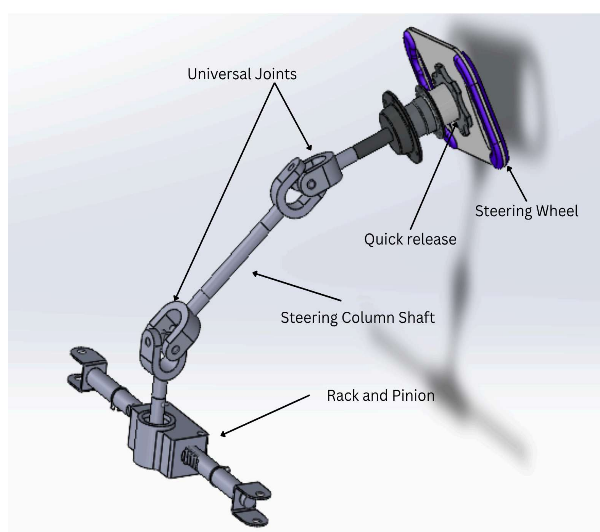
2.Driver comfort is ensured: Required hand force (~125 N) is suitable for short-duration racing and offers good feedback.

3.High friction and scrub radius were considered: Worst-case scenario was simulated (static friction = 1.4), so dynamic effort will be even lower.

4.Compact steering ratio improves responsiveness: A ratio of 3.2 allows quick steering input with reasonable mechanical advantage.

5.Optimized for packaging and ergonomics: 240 mm steering wheel balances control and cockpit space, without exceeding force limits.

CAD Model:



DATA FOR STEERING RESPONSE IN LOTUS STIMULATION:

FRONT SUSPENSION - STEERING TRAVEL

TYPE 14 Double Wishbone, Push Rod to damper

INCREMENTAL GEOMETRY VALUES

RACK TRAVEL (mm)	TOE ANGLE RHS (deg)	TOE ANGLE LHS (deg)	CAMBER ANGLE RHS (deg)	CAMBER ANGLE LHS (deg)	ACKERMANN (%)	TURNING CIRCLE RADIUS (mm)
-40.00	-26.17	46.92	-1.58	4.22	-100.31	2302.14
-35.00	-23.10	44.60	-1.56	3.58	-119.85	2603.02
-30.00	-20.00	30.49	-1.52	1.21	-104.18	3445.67
-25.00	-16.86	22.83	-1.47	0.40	-96.28	4398.21
-20.00	-13.67	17.08	-1.41	-0.06	-92.17	5709.50
-15.00	-10.41	12.19	-1.34	-0.39	-89.76	7807.08
-10.00	-7.06	7.82	-1.24	-0.64	-88.37	11904.63
-5.00	-3.60	3.78	-1.13	-0.84	-87.67	24032.74
0.00	0.00	0.00	-1.00	-1.00	-100.31	0.00
5.00	3.78	-3.60	-0.84	-1.13	-87.67	24032.74
10.00	7.82	-7.06	-0.64	-1.24	-88.37	11904.63
15.00	12.19	-10.41	-0.39	-1.34	-89.76	7807.08
20.00	17.08	-13.67	-0.06	-1.41	-92.17	5709.50
25.00	22.83	-16.86	0.40	-1.47	-96.28	4398.21
30.00	30.49	-20.00	1.21	-1.52	-104.18	3445.67
35.00	44.60	-23.10	3.58	-1.56	-119.85	2603.02
40.00	46.92	-26.17	4.22	-1.58	-100.31	2302.14

★ **CONCLUSION:**The complete vehicle design has been developed with a focus on performance, safety, and reliability, aligning with FSAE standards. Every subsystem—from steering and suspension to powertrain and ergonomics—has been optimized using analytical calculations, simulations, and practical constraints. The final design ensures efficient driver control, structural integrity, and competitive dynamics on track. All design choices have been validated with sound engineering principles, ensuring that the car is ready for manufacturing and testing with confidence in its performance and handling characteristics.

- **Steering wheel:**

Material	Aluminium-7075-T6
Manufacturing process	Waterjet
No	1

Fig: Steering Wheel

- **Quick release mechanism:**

Material	Aluminium-7075-T6
Manufacturing process	OEM
No	1

Fig: Quick release

- **Steering spline:**

Material	En-8
Manufacturing process	OEM
No	1

Fig: Steering spline

- **Universal joint:**

Material	Alloy Steel
Manufacturing process	OEM
No	2

Fig: Universal joint

- **Steering column:**

Material	En-8 Rod
Manufacturing process	OEM
No	1

Fig: Steering Column

- **Rack and Pinion:**

Manufacturing process	Custom manufacturing
No	1